

PROTEIN STRUCTURE PREDICTION REPORT:

Human Insulin

Protein Details: Human insulin is a 110 amino acid polypeptide hormone essential for glucose metabolism. The protein exhibits a compact, globular fold stabilized by disulfide bonds.

Structure Prediction Method: The 3D structure was predicted using SWISS-MODEL based on the template 7sth.1.C (insulin receptor bound structure) with 100% sequence identity coverage. The predicted model achieved a QMEAN score of 0.26, indicating a reliable structural prediction.

Key Structural Features: The ribbon diagram reveals insulin's characteristic fold composed of alpha-helices (blue regions) and extended loops (orange regions). The spacefill representation demonstrates the tight molecular packing, with the core hydrophobic residues forming the central domain and polar residues on the surface. The structure exhibits two distinct domains connected by flexible linker regions, typical of peptide hormone architecture.

Functional Significance: The predicted 3D structure highlights conserved structural motifs critical for receptor binding and biological activity. The presence of multiple secondary structural elements suggests high stability and specificity for insulin receptor interactions.

